

Varun Kannan

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Summary

Quantitative researcher with a theoretical physics background focused on the modeling of noisy systems. Creates scalable portfolios with the use of mathematical tools from theoretical physics, with current work on mathematical modeling of stocks, ETFs, and options in differing market regimes.

Education

2018-09 - 2022-06: Bachelor of Science: Physics, Minor in Mathematics

University of California - San Diego - La Jolla, CA

GPA: 3.32 | 3.60 Major

Honors: Provost Honors (5 terms), Salutatorian (HS)

Technical Skills

Programming

- Python (Pandas, NumPy, Matplotlib/Plotly, VectorBt, DuckDB, Multiprocessing)
- Java, C#, JavaScript, SQL, Mathematica

Tools

- PowerShell, Linux, Software IDEs, JSON/Parquet data pipelines

Projects

Quantitative Trading Model 06/2025 - Current (11 months)

- Python-based backtesting framework built for research purposes and general usability (GitHub: <https://github.com/sponge-mi-boi/Quant-data>)
- Designing Trading Strategies with accurate signal extraction, model validation, and scalability analysis on low/medium frequency timescales
- Evaluated with the use of risk-adjusted performance metrics
- Vectorized research pipeline focused on efficient data extraction and the use of walk-forward analysis to validate both out of sample performance and multi-regime stability
- Testing of various quantitative methods including PCA-based factor models, multi-stock cointegration, HMMs, and statistical arbitrage based portfolios
- Results/methodology is being documented publicly: [Quant Finance](#)

Technical Experience

Quantum Physics Research

Internship (Paid)/Honors Research: 06/2021 - 12/2021 (6 months) UCSD Physics Department, La Jolla, CA

Dr. Daniel Green, Associate Professor at UCSD

- Identified issues in existing models and tested proposed models on existing data with rigorous analysis tools and a comprehensive research process. Use of Bayesian filters to estimate hidden variables in the context of quantum phenomena, such as decoherence. Analyzed research data from various technical sources using Mathematica to create representative graphs, highlighting key insights.

Independent Study/Research 03/2021 - 12/2021 (9 months) UCSD Physics Department, La Jolla, CA

Dr. George Fuller, Distinguished Professor, Former Director of the Center for Astrophysics and Space Science, UCSD

- A mentorship on various topics in the field of General Relativity. Proposed first principles based models that agree with existing data. Obtained testable predictions to analyze possible sources of error.

Associate Researcher 10/2024 - 06/2025 (8 months) UCSD Math Department, La Jolla, CA

- Performed fundamental research to author a technical paper based on rigorous mathematical analysis and existing physics-based frameworks.

Tutor 05/2025 - Current (11 months) Remote/Several

- Private tutor in mathematics and physics for undergraduate level/ high school students

Publications and Technical Writing

- Authored a paper based on individual study along with a review from a distinguished professor
- Created the website, <https://www.qgspinor.com>, where I post technical writing, short review articles, derivations, and related exploratory topics.

References

References available upon request